# Mohamed H. Essa

## Flutter Developer — Enterprise Mobile Applications

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## SUMMARY

Flutter developer with 3+ years of experience (2 years freelance, 2 years in-house) building enterprise mobile applications, including systems for Egypt's Ministry of Health (600+ healthcare facilities) and UNICEF, and e-commerce platforms for clients in the UAE and Oman. Strong in Clean Architecture, BLoC/Cubit and Riverpod state management, offline-first data sync, and dynamic form workflows. Backend-capable (NestJS/TypeScript, Laravel and Python maintenance) with verified open-source contributions to the Nilefy low-code platform. AWS Certified Solutions Architect – Associate.

## SKILLS

**Mobile:** Flutter, Dart, BLoC/Cubit, Riverpod, Clean Architecture, offline-first sync, dynamic forms & validation, Dio/Retrofit, Hive & secure storage, Firebase (Auth, FCM, Firestore), push notifications, geolocation, biometrics, widget & unit testing

**Backend:** NestJS, TypeScript, Node.js, Laravel (maintenance), Python, PostgreSQL, REST API design, Drizzle ORM

**Cloud & delivery (supporting):** AWS (certified SAA-C03), Docker, CI/CD with GitHub Actions & Codemagic, Git; home lab running Kubernetes, PostgreSQL and Grafana/Prometheus

**Languages:** Arabic (native), English (fluent), German (learning)

## EXPERIENCE

### Flutter Developer — Intrazero, Cairo, Egypt

*Aug 2024 – Present*

- Built Egypt's National Medical Waste Management System for the Ministry of Health as a Flutter application digitizing waste tracking across **600+ healthcare facilities**, with multi-level approval workflows.
- Designed a reusable form-orchestration engine (typed generic inputs, cross-field validation, map-driven rendering) powering form-heavy government workflows; a public counterpart of the pattern is open-sourced on my GitHub.
- Developed core features for a UNICEF child-protection case-management platform, including offline-first delayed sync and dynamic forms for field workers.
- Delivered and maintained e-commerce applications for clients in the UAE and Oman (Motary, Al-Diplomacy), covering multi-language support, product search, and regional payment integrations.
- Maintained and fixed backend services in Laravel and Python supporting the mobile products.

### Open-Source Contributor — Nilefy (low-code platform, NestJS/React)

*Oct 2023 – Jun 2024 · github.com/nilefy/nilefy*

- Contributed 118 commits (~12,700 lines) to the platform's TypeScript/NestJS backend.
- Implemented the encryption/decryption module securing data-source credentials, including nested-configuration handling.
- Built the application export/import pipeline, including component-tree serialization and DTO validation.
- Developed the REST API data-source connector with custom-header support, and table CRUD operations with transactional Drizzle ORM logic.
- Extended authentication with forgot-password and email-confirmation flows across backend and React frontend.

**Freelance Mobile & Backend Developer — Remote**

*Jun 2022 – Jul 2024*

- Delivered cross-platform client solutions using Flutter for mobile and Node.js/NestJS with Firebase and PostgreSQL for backends.
- Owned projects end-to-end: requirements, architecture, implementation, release, and post-launch maintenance.

## SELECTED PROJECTS (public code — github.com/Mohamed-H-Essa)

- **star_wars_land** — Clean Architecture reference app: BLoC, offline caching, 30 test files, CI via GitHub Actions.
- **flutter_form_engine** — reusable dynamic-form engine with typed generic inputs and sync/async validation (Formz + BLoC).
- **wfm** — production-scale workforce-management app: Riverpod 2, Dio/Retrofit, Hive + secure storage, biometric auth, FCM.
- **Slipway** (k8s_platform_bootstrap) — a production-style Kubernetes platform that runs on a laptop: Terraform, Argo CD GitOps, signed images enforced by Kyverno, SLO alerts, and a chaos day under load.

## CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate (SAA-C03)** — score 832/1000, issued Jun 2026 (Credly-verified)

## EDUCATION & COMMUNITY

- **B.Sc. Computer Engineering, Benha University** — Cairo, Egypt, 2019 – 2024
- Flutter Instructor (volunteer), IEEE Student Branch, Benha University — designed and taught a Flutter curriculum for engineering students